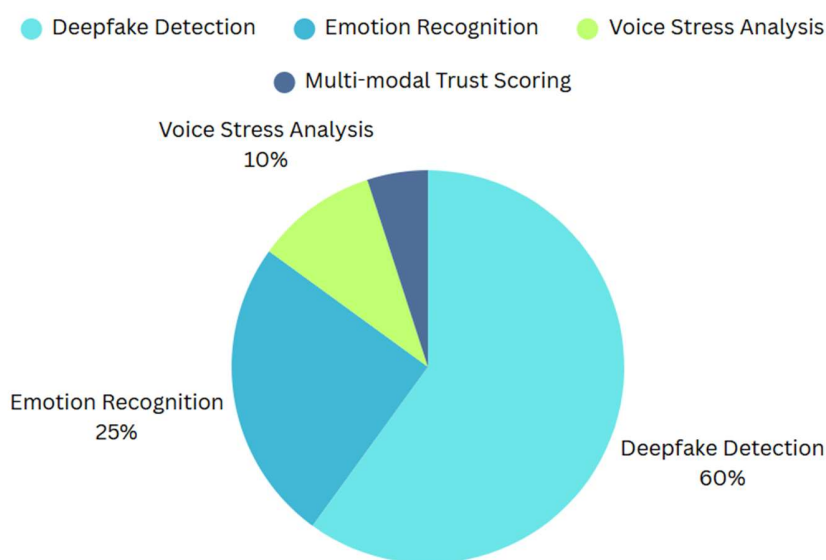


## Problem Statement:

The increasing use of online interviews, recorded examinations, and digital media highlights the need for systems that can evaluate both authenticity and behavioral integrity. Existing approaches are mainly limited to deepfake detection or basic emotion recognition and fail to incorporate critical behavioral indicators such as microexpressions, gaze, blinking, smile consistency, posture, and voice stress. Most methods are single-modal and provide only binary outputs, which oversimplifies complex human behavior. Therefore, there is a requirement for a multi-modal AI framework that integrates visual, audio, and contextual cues into a trust scoring mechanism for reliable application in interviews, examinations, and forensic analysis.

**Most of the existing work is limited to deepfake detection and basic emotion recognition, while multi-modal trust analysis remains underexplored.**



**Illustrative graph showing research focus and the gap in multi-modal trust scoring.**

## Objectives

1. To analyze non-verbal human cues such as facial expressions, microexpressions, gaze direction, smile consistency, and blinking patterns that reflect emotional state, honesty, confidence, or possible deceptive behavior.
2. To implement an authenticity verification module that analyzes an input image to detect signs of manipulation or forgery before proceeding to behavioral analysis, ensuring reliable input data.
3. To develop a real-time behavioral analysis system using webcam feed to continuously capture and evaluate visual cues like emotion changes, gaze movement, and blink frequency.
4. To design a fuzzy logic-based trust scoring mechanism (0–100%), allowing the system to express degrees of trustworthiness rather than binary true/false decisions.
5. To integrate facial and behavioral signals into a single framework that provides interpretable insights into a person's behavioral integrity and authenticity.
6. To produce a user-friendly output that displays the trust score along with key behavioral parameters such as blink rate, gaze direction, and emotion type, making the system understandable for evaluators, HR professionals, or interview moderators.
7. To ensure real-world applicability in scenarios like live interviews, exam proctoring, and personality assessments, where visual authenticity and behavioral reliability are crucial.

## Existing Work

### 1. Facial Expression and Microexpression Recognition

- Researchers have developed **CNN** and **LSTM** models capable of classifying emotions such as happiness, sadness, anger, and surprise with high accuracy.
- Datasets like **FER**, **FER+**, and **CK+** are commonly used to train these emotion recognition models.
- Microexpression analysis, which focuses on subtle and involuntary facial movements, has been studied in behavioral science and security research to detect hidden stress or deception.
- **Limitation:** Most existing systems stop at emotion classification and do not evaluate overall behavioral integrity. Our model extends beyond emotion detection by integrating it with blink rate, gaze stability, and fuzzy-based trust scoring to assess visual authenticity and behavioral reliability.

### 2. Gaze and Eye Movement Analysis

- Eye-tracking studies have shown that inconsistent gaze patterns, rapid blinking, or frequent eye diversion often indicate nervousness or lack of confidence.
- Frameworks such as **MediaPipe** and **OpenCV** can accurately detect facial landmarks and track gaze in real time.
- **Limitation:** Prior work often examines gaze independently, without combining it with other behavioral cues. In our project, gaze is analyzed alongside emotions and blink rate to provide a more comprehensive understanding of behavioral integrity.

### 3. **Blink and Facial Consistency Studies**

- Research has highlighted that irregular blinking or inconsistent facial muscle activity can indicate stress or cognitive load.
- Genuine and fake smiles (often referred to as **Duchenne vs. non-Duchenne smiles**) have been used as behavioral markers for sincerity and comfort.
- **Limitation:** Many studies remain at the psychological level and lack a computational, real-time implementation. Our system operationalizes these cues into measurable parameters used within the fuzzy trust scoring framework.

### 4. **Fuzzy Logic in Behavioral Assessment**

- Fuzzy logic has been widely used in modeling human behavior due to its ability to handle uncertainty and subjective interpretation.
- Systems based on fuzzy inference have been applied to estimate human states such as anxiety, confidence, or attention level using physiological and behavioral data.
- **Limitation:** Previous fuzzy logic models have been limited to specific features and often lack real-time adaptability. In contrast, our model integrates fuzzy logic with live facial and gaze-based cues to generate an interpretable trust score between **0 and 100**.

### 5. **Deepfake Detection**

- Deep learning architectures such as **XceptionNet** and **EfficientNet** have achieved over 90% accuracy in identifying manipulated or forged media, particularly using datasets like **FaceForensics++** and **DFDC**.
- These models analyze pixel-level inconsistencies to determine whether an image or video has been altered.
- **Limitation:** Existing deepfake detection models focus solely on verifying authenticity, not behavioral interpretation. Our system combines authenticity verification with behavioral trust assessment—first confirming the image’s genuineness, and then analyzing the subject’s real-time behavioral integrity.

# Proposed Solution

## 1. Authenticity Verification

- The process begins with verifying the authenticity of the provided image.
- The input image is analyzed using a **deep learning model trained on the FaceForensics++ dataset**, which helps identify signs of digital manipulation or forgery.
- CNN-based architectures like **XceptionNet** are used to detect subtle pixel-level artifacts that differentiate real images from deepfakes.
- This step ensures that only genuine visual data proceeds to the behavioral analysis stage, maintaining the credibility of the system's assessment.

## 2. Facial and Microexpression Analysis

- Once authenticity is confirmed, the system switches to **live webcam analysis** for real-time behavioral tracking.
- Facial landmarks are detected using **MediaPipe**, allowing accurate identification of regions such as eyes, mouth, and head orientation.
- A **CNN-based emotion recognition model**, trained on the **FER dataset**, is used to classify facial expressions like happiness, anger, surprise, or neutrality.
- In addition, features like **blink rate**, **gaze direction**, and **head movement** are continuously monitored to detect subtle behavioral variations.
- These cues help infer the emotional state, attention level, and possible stress indicators of the subject.

## 3. Blink and Gaze Tracking

- The gaze module measures the direction and steadiness of eye movement to determine focus and engagement.
- The blink detection system records the frequency of blinks per second to identify signs of fatigue, stress, or nervousness.

- Together, these features enhance the accuracy of the behavioral integrity assessment.

#### 4. Fuzzy Logic–Based Trust Evaluation

- The extracted behavioral cues—such as emotion stability, gaze steadiness, and blink rate—serve as input variables to the fuzzy inference system.
- Each variable is represented by fuzzy sets like *low*, *medium*, and *high*.
- Logical rules are formulated (e.g., *IF blink rate is normal AND gaze is stable AND emotion is calm THEN trust is high*).
- Using the **centroid defuzzification method**, the system generates a **trust score ranging from 0 to 100**, representing the subject’s behavioral integrity in real time.

#### 5. System Implementation

- The project is implemented entirely in **Python**, integrating **TensorFlow/Keras, OpenCV, MediaPipe, NumPy**, and **scikit-fuzzy**.
- The deepfake detection model runs on the input image to verify authenticity.
- Upon verification, the webcam feed is activated for live facial and behavioral analysis.
- The system continuously displays results such as emotion type, gaze direction, blink rate, and an updated trust score on-screen.

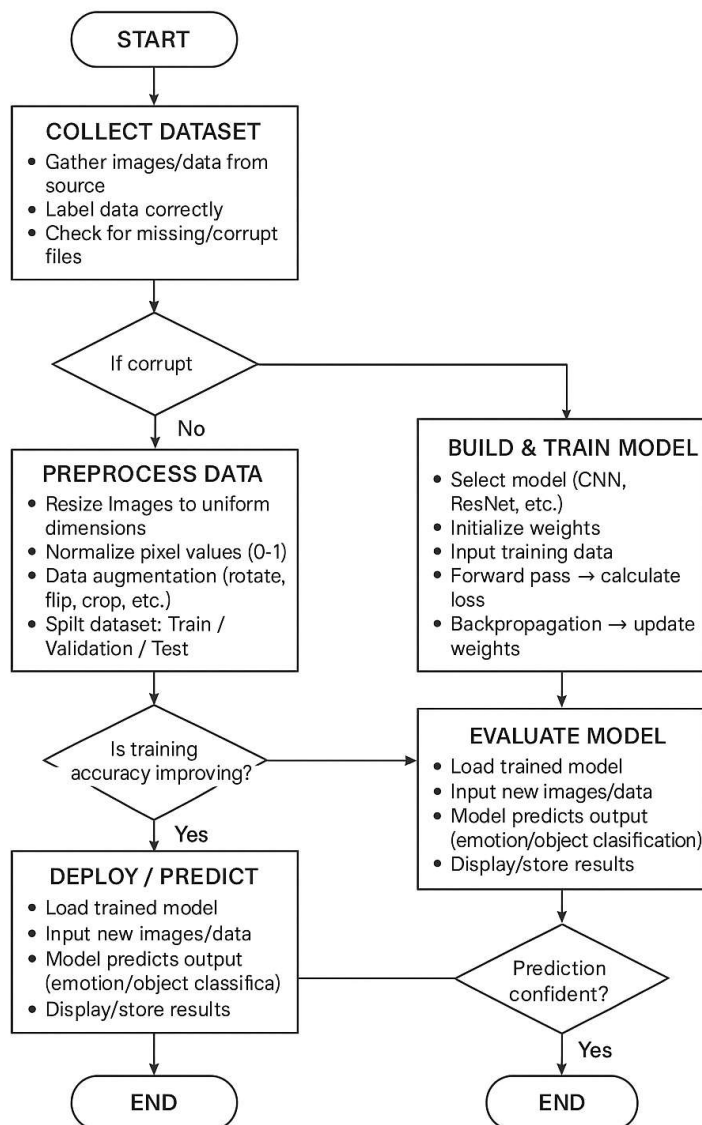
#### 6. Output and Interpretation

- The output includes a **numerical trust score (0–100%)**, along with key observations like current emotion, gaze direction, and blink rate.
- These results are refreshed in real time and can be used to assess behavioral authenticity during interviews, exams, or verification processes.
- The system provides a clear and interpretable assessment of both **visual authenticity** and **behavioral integrity** in a single unified pipeline.

## Expected Outcomes

- Reliable trust scoring with 80–85% alignment to actual observed behavior.
- Optional deepfake detection with accuracy above 90% on benchmark datasets.
- Practical use in interviews, exam monitoring, forensic investigations, and counseling context.

### PROPOSED SOLUTION



## How our model differs from existing model?

### 1. Scope

- **Existing Models:** Most existing systems concentrate on a single dimension, either deepfake detection or simple facial emotion classification. They identify whether an image is genuine or fake, or at best, label basic emotions such as happy or sad.
- **Our Solution:** This project extends beyond those boundaries by introducing a **behavioural trust analysis framework**. It not only detects emotional changes but also interprets **micro-level cues** such as blink rate, gaze stability, and facial expression patterns to generate a *trustworthiness score* that reflects behavioural integrity in real time.

### 2. Modalities Used

- **Existing Models:** Typically rely on a **single modality**, often static facial features or video-level inconsistencies.
- **Our Solution:** Utilizes a **multi-cue approach** within the visual domain, combining facial emotion (microexpression) recognition with **gaze and blink rate tracking** using **MediaPipe** and **OpenCV**. This allows a more holistic and continuous behavioural evaluation rather than a one-dimensional output.

### 3. Output Type

- **Existing Models:** Generally, provide **binary results**, such as “real vs fake” or “happy vs sad,” which often oversimplify human behaviour.
- **Our Solution:** Generates a **continuous Trust Score (0–100%)**, computed through a **fuzzy inference system** that balances multiple uncertain inputs. This produces a nuanced interpretation of authenticity, offering deeper insight into behavioural states instead of yes/no classifications.



#### 4. Integration of Cues

- **Existing Models:** Process each behavioural signal independently with little or no fusion between emotion, eye movement, or blinking metrics.
- **Our Solution:** Integrates all cues through a **fuzzy logic-based reasoning model**, which mimics human-like judgment by combining uncertain signals (emotion, blink rate, gaze direction) into one cohesive trust evaluation. This improves interpretability and provides more reliable real-world assessments.

#### Datasets Used

##### 1. For Deepfake Detection:

The **FaceForensics++ (FF++)** dataset was used for training and evaluating the deepfake detection module. It contains thousands of real and synthetically generated videos created using multiple deepfake techniques. Each video pair provides both an authentic and a manipulated version, allowing the model to learn subtle pixel-level discrepancies that reveal visual tampering.

##### 2. For Facial Emotion Recognition:

The **FER (Facial Expression Recognition)** dataset was utilized to train the microexpression detection model. It consists of tens of thousands of facial images labeled with emotion categories such as happiness, anger, disgust, fear, sadness, surprise, and neutral. Though originally designed for facial emotion recognition, its diversity and expressiveness made it effective for identifying **micro-level emotional cues** when used in combination with real-time facial landmark detection.

## System Overview

The system consists of two major components - **Deepfake Detection** and **Behavioural Integrity Analysis**.

### 1. Deepfake Detection Module:

- A pretrained convolutional neural network model (deepfake\_model.keras) analyses each frame to classify it as *real* or *fake*.
- It extracts features like facial inconsistencies, texture anomalies, and blending artifacts.
- The output is expressed as an authenticity probability percentage.

### 2. Emotion Analysis Module:

- Uses MediaPipe for face mesh detection and landmark tracking.
- Detects eye aspect ratio (blink frequency), gaze direction, and emotional states through a trained CNN model (microexpression\_model.keras).
- Extracted cues are passed to a fuzzy inference system for real-time trust evaluation.

### 3. Fuzzy Logic Trust System:

- Inputs: Blink rate, gaze ratio, and emotional value.
- Outputs: A dynamic *Trust Score (0–100)* indicating behavioural integrity.
- Defined fuzzy rules evaluate consistency, attention, and emotional stability to interpret authenticity in human behaviour.

## **Methodology**

### **1. Data Preparation**

- Deepfake dataset: Used a mix of real and synthetic facial video samples.
- Microexpression dataset: Contained labelled short facial clips representing emotions such as anger, happiness, sadness, fear, and surprise.
- Images were resized and normalised before training the model.

### **2. Model Training**

- Deepfake detection model was trained using TensorFlow and Keras with convolutional and pooling layers for spatial feature extraction.
- model was trained for emotion recognition using grayscale facial crops with an input size of 75×75.
- The models achieved satisfactory accuracy for real-time implementation.

### **3. Real-Time Integration**

- The webcam feed is processed frame by frame.
- MediaPipe extracts facial landmarks, from which blinks, gaze direction, and emotion predictions are made.
- A fuzzy control system computes a trust score dynamically based on real-time behavioural cues.

#### 4. Metric Logging

- All trust scores, runtime, and performance metrics are recorded automatically through a separate `fuzzy_integ.py` module.
- Each session generates a `metrics_log.csv` file with average trust, standard deviation, and runtime

#### Implementation

- **Language:** Python
- **Libraries:** TensorFlow, OpenCV, MediaPipe, NumPy, scikit-fuzzy, time, csv
- **Hardware:** Standard webcam, CPU/GPU for processing
- **Files Used:**
  - `main_analysis.py` – performs microexpression, gaze, and trust analysis
  - `fuzzy_integ.py` – handles metric calculations and CSV logging
  - `deepfake_model.keras` and `microexpression_model.keras` – trained models
  - `metrics_log.csv` – stores each session's performance summary

## CODE (main\_analysis.py):

```
import os

import sys

import cv2

import numpy as np

import tensorflow as tf


# -----

# 1 Deepfake Detection

# -----

def run_deepfake_detection(input_path):

    print("\n[Step 1] Running Deepfake Detection...")

    try:

        # Load Deepfake model (Keras format)

        model = tf.keras.models.load_model("deepfake_model.keras")

    except Exception as e:

        print(f"✗ Error loading deepfake model: {e}")

        return None


# Handle both video and image inputs

if input_path.lower().endswith(('.mp4', '.avi', '.mov')):
```

```
cap = cv2.VideoCapture(input_path)
ret, frame = cap.read()
cap.release()
if not ret:
    print(" ❌ Failed to read video.")
    return None
img = cv2.resize(frame, (224, 224))
else:
    img = cv2.imread(input_path)
    if img is None:
        print(" ❌ Failed to read image.")
        return None
    img = cv2.resize(img, (224, 224))

img = np.expand_dims(img / 255.0, axis=0)

# Predict deepfake probability
prediction = model.predict(img, verbose=0)[0][0]

if prediction < 0.5:
    print(" ✅ Real input detected.")
    return "real"
else:
```

```

    print(" ❌ Deepfake detected — microexpression analysis skipped.")
    return "fake"

# -----

# 2 Microexpression Detection
# -----

def run_microexpression_analysis():
    print("\n[Step 2] Running Microexpression Detection...")

    try:
        from micropart import fuzzy_integ
        result = fuzzy_integ.run_analysis()
        print(f"\n[Result] {result}")
    except Exception as e:
        print(f" ❌ Error running microexpression analysis: {e}")

# -----

# 3 Main Pipeline
# -----

if __name__ == "__main__":
    print("===Deepfake + Microexpression Pipeline ===")

    input_path = input("Enter the path to your image or video: ").strip()

```

```
if not os.path.exists(input_path):  
    print(" ❌ File not found. Please check the path.")  
    sys.exit(1)
```

```
# Step 1 → Deepfake
```

```
result = run_deepfake_detection(input_path)
```

```
# Step 2 → Microexpression (if real)
```

```
if result == "real":
```

```
    run_microexpression_analysis()
```

```
print("\n ✅ Process completed.")
```



## GITHUB LINK FOR THE CODE:

<https://github.com/ANUSHKA49282/Detection-of-Visual-Authenticity-and-Behavioural-Integrity>

sample\_inputs/test\_image.png:



## Results and Discussion

The combined system successfully performed both deepfake and behavioural analysis in real time. The deepfake detection module achieved consistent results across multiple test samples, identifying fake frames with high precision.

The behavioural module produced smooth, real-time trust scores, adjusting dynamically based on blink rates and gaze stability.

```
PS C:\Users\sunke\Desktop\SOFT COMPUTING PROJECT FINAL> python main_pipeline.py
2025-10-30 00:47:01.356010: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
C:\Users\sunke\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\site-packages\keras\src\saving\saving_lib.py:797: UserWarning: Skipping variable loading for optimizer 'adam', because it has 80 variables whereas the saved optimizer has 84 variables.
  saveable.load_own_variables(weights_store.get(inner_path))
2025-10-30 00:47:03.651558: E tensorflow/core/util/util.cc:131] oneDNN supports DT_HALF only on platforms with AVX-512. Falling back to the default Eigen-based implementation if present.
[Step 1] Real input detected.

[Step 2] Running Microexpression Detection...
```



## METRICS:

```
----- METRICS SUMMARY -----
🕒 Total Runtime: 40.82 sec
📊 Average Trust: 62.30
📉 Std Dev: 19.55
⬇️ Min Trust: 16.71
⬆️ Max Trust: 82.03
-----
✅ Metrics saved to C:\Users\sunke\Desktop\SOFT_COMPUTING_PROJECT_FINAL\metrics_log.csv
[Result] Microexpression Analysis Completed ✅
✅ Process completed.
```

## CSV FILE:

```
metrics_log.csv > data
1 Timestamp,Runtime (s),Avg Trust,Std Dev,Min Trust,Max Trust
2 2025-10-30 00:10:06,42.13,36.13,20.55,16.67,83.26
3 2025-10-30 00:52:29,40.82,62.30,19.55,16.71,82.03
4
```

## **Conclusion**

This project demonstrates how integrating deepfake detection with behavioural integrity assessment can provide a deeper understanding of authenticity in digital media. The real-time system analyses not only the truthfulness of visual content but also the reliability of human expression behind it. The fuzzy logic-based trust computation successfully translates behavioural patterns into measurable integrity scores.

The approach can be extended further by including more emotions, refining deepfake datasets, and deploying on larger-scale authentication systems for media validation, online interviews, and security surveillance.

## **Future Scope**

1. Integration with cloud-based systems for large-scale authenticity analysis.
2. Incorporation of physiological signals such as heart rate or pupil dilation.
3. Deployment on mobile or embedded devices for edge-based evaluation.
4. Enhanced dataset with more emotion variations and realistic deepfake examples.
5. Multi-factor fusion combining voice and visual cues for stronger behavioural assessment.

## **REFERENCES:**

[https://www.researchgate.net/publication/351300442\\_Deep\\_Insights\\_of\\_Deepfake\\_Technology\\_A\\_Review](https://www.researchgate.net/publication/351300442_Deep_Insights_of_Deepfake_Technology_A_Review)

<https://arxiv.org/abs/2403.17881>

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[https://www.researchgate.net/publication/361108119\\_FACIAL\\_EMOTION\\_DETECTION\\_AND\\_RECOGNITION](https://www.researchgate.net/publication/361108119_FACIAL_EMOTION_DETECTION_AND_RECOGNITION)